

# Deep Learning School

## Visual Autoregressive Models



- Predicts pixels iteratively in sequence
- Creates detailed and realistic images
- Useful for generative AI applications

# Image generation

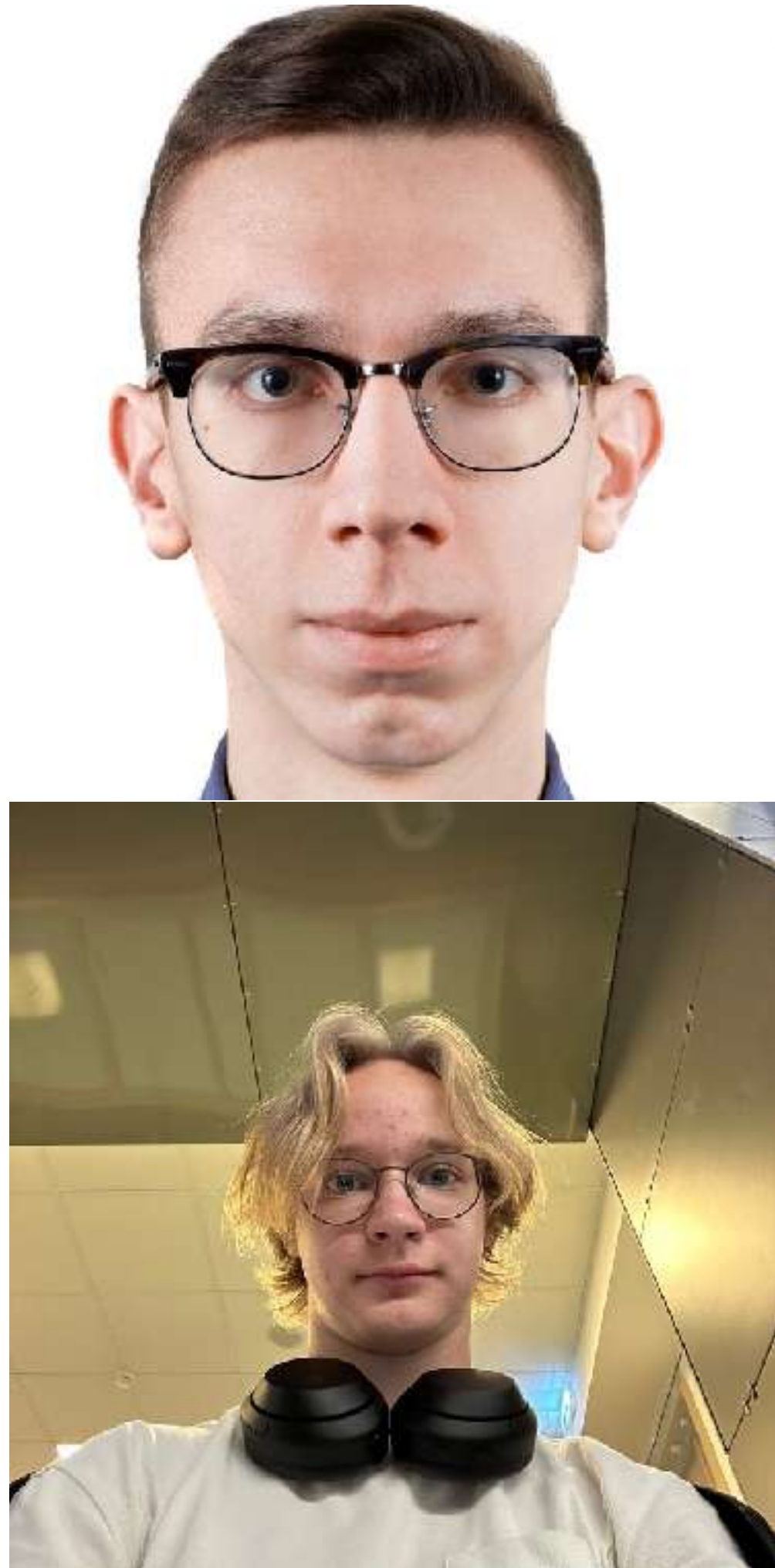


*A beautiful futuristic world with intelligent dinosaurs*



*Heroes of might and magic V; inferno vs necropolis fight; 100 weeks*

# Image editing



# Video generation

*Animate the people in these images so they come to life and step out of the photos...*

*Preserve their facial features, expressions, and identity as accurately as possible.*

*Transform each person into a different animal. After the transformation, place them inside a luxurious Bugatti Veyron.*

*Show them seated naturally as if on a road trip. Then create a cinematic sequence of the car driving upward, leaving Earth and traveling through space toward the Moon, with dramatic lighting, stars, and planetary scenery in the background.*



# World generation



*A vibrant 3D style, an adorable, fluffy creature bounding across a vibrant rainbow bridge in a fantastical landscape.*

*The rainbow bridge arches gracefully through a whimsical landscape, perhaps filled with floating islands, glowing flora, and swirling clouds.*

...

**Diffusion-based models is a leading paradigm for visual generation**



**Diffusion**



**AR models**

# Why AR visual models?

**Making use of pretrained LLMs:** even text-text pretrain is useful for visual generation

**Not AR vs Diffusion**, but **AR + Diffusion**



e.g.

**Long video generation:** diffusion for spatial dimensions and AR for temporal

# Outline

**Goal for today:** understanding visual AR basics

01

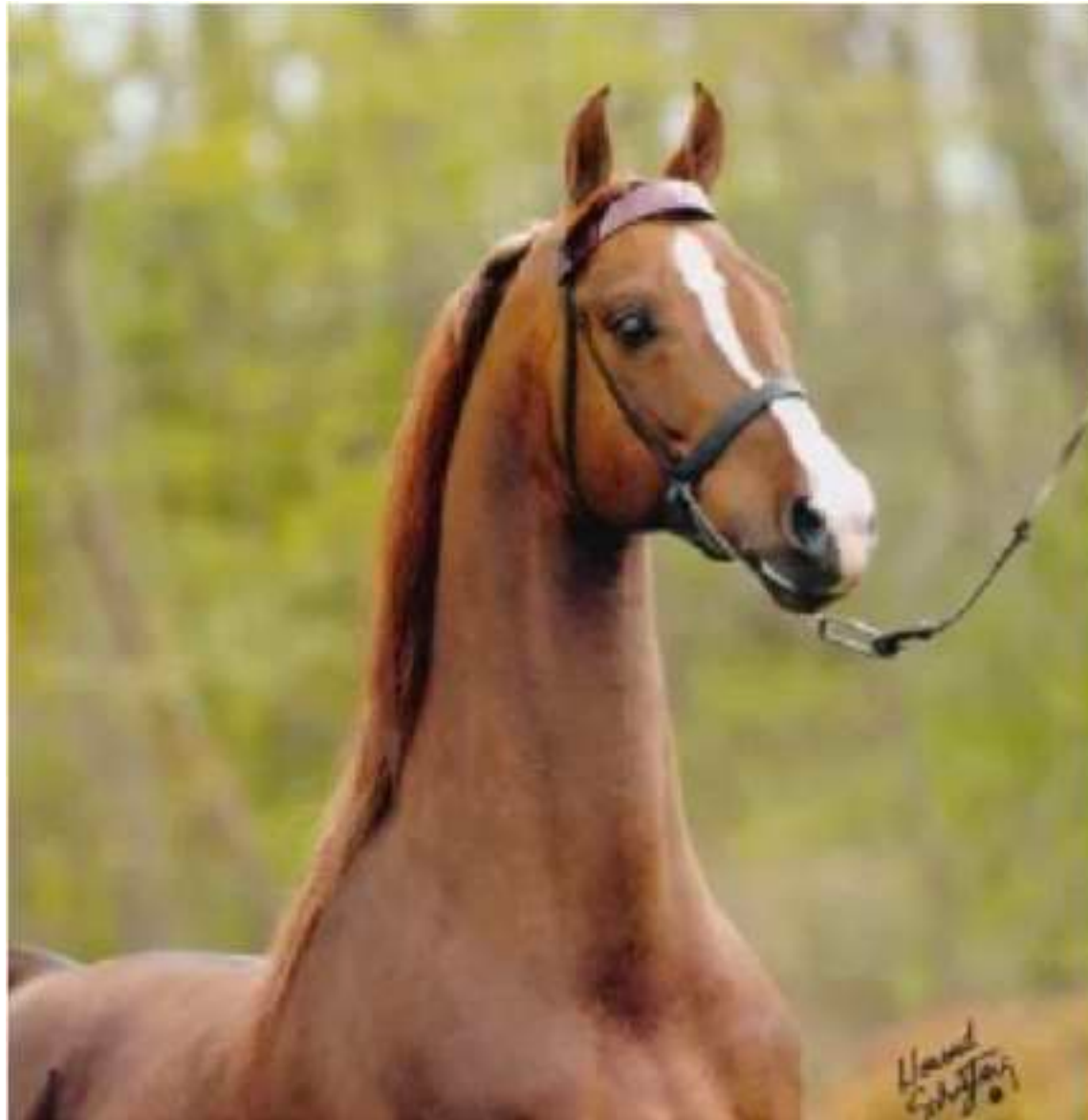
**Classical visual AR modeling**

02

**Next-scale prediction modeling**

# Image tokenization

Similar to LLMs, we first need to represent **images as a sequence of discrete tokens**



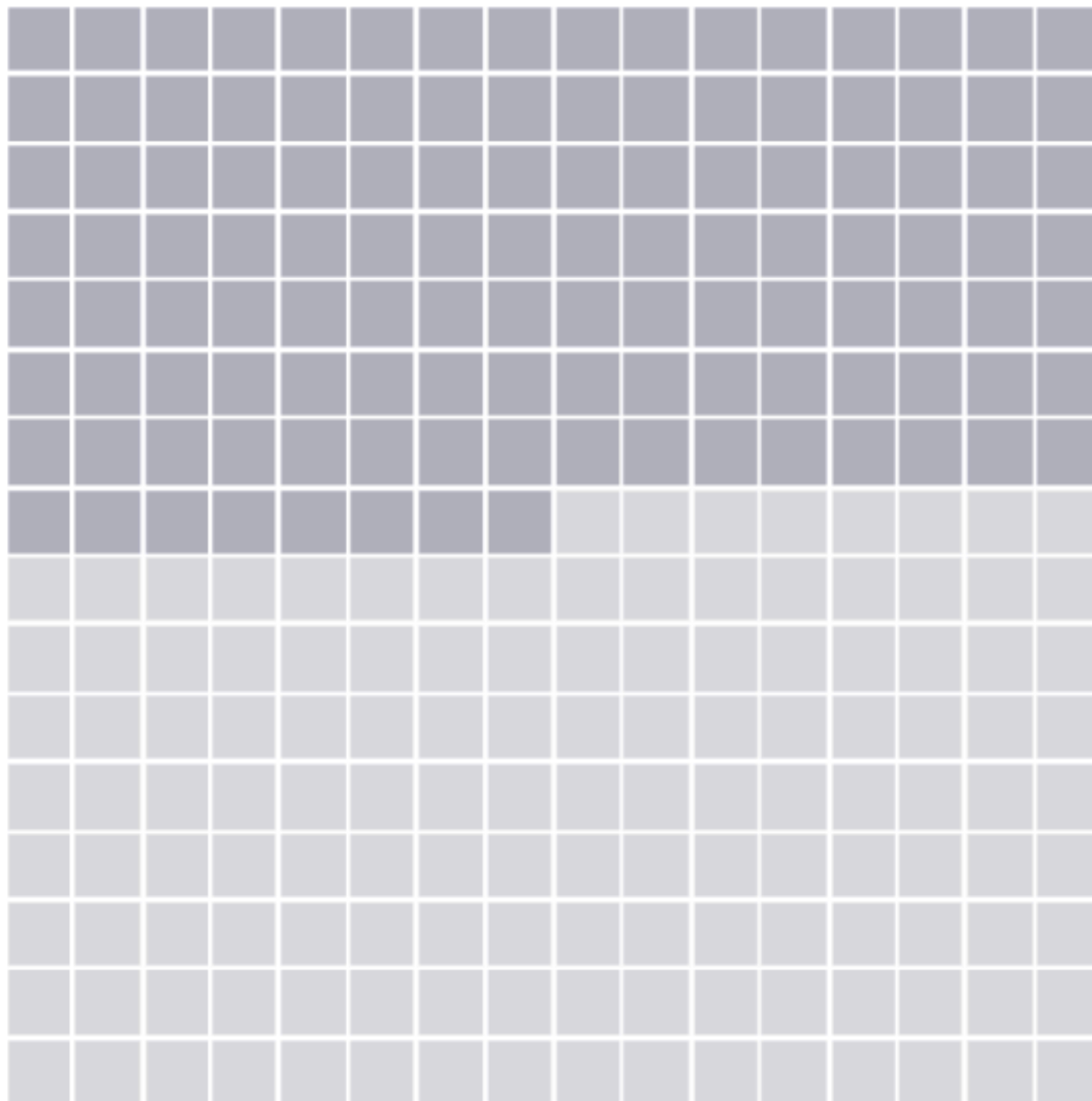
Discrete tokens

# Naive image tokenization

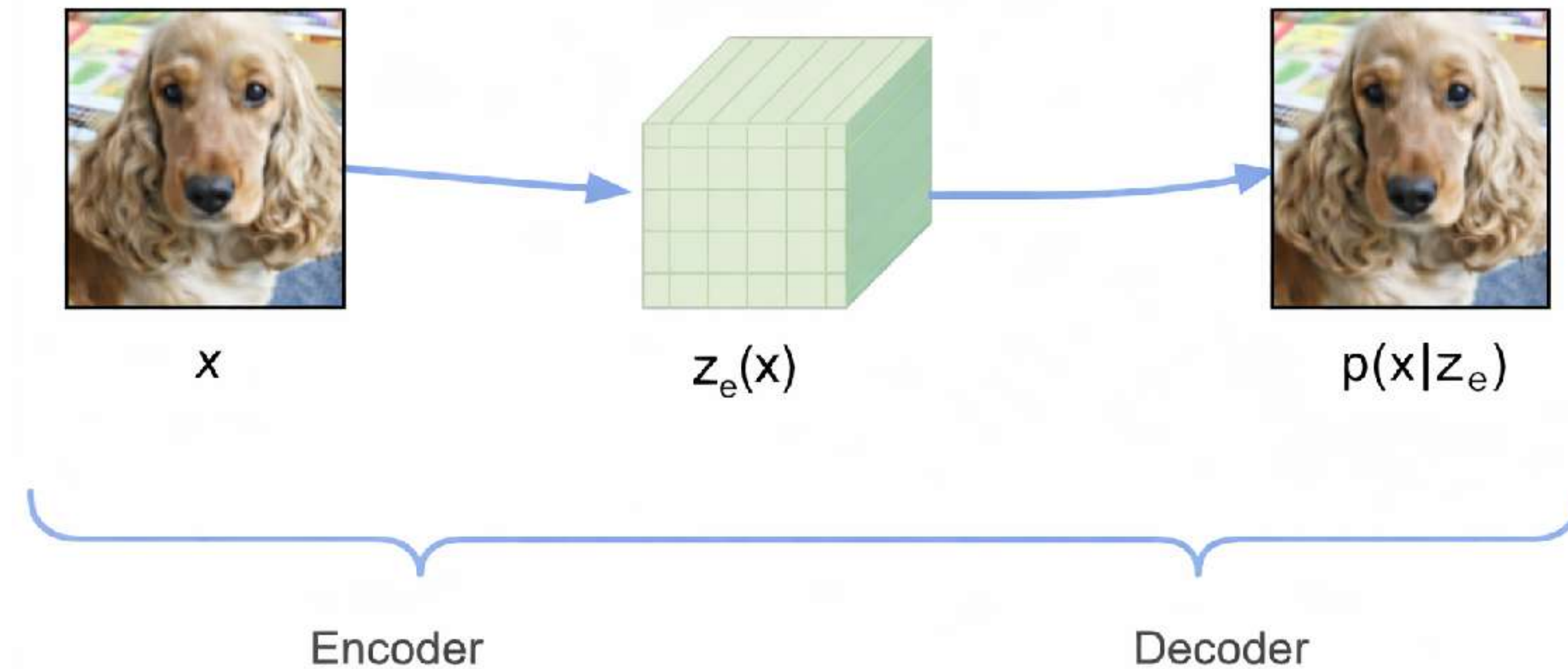
**Pixels:** each pixels is a discrete value in  $[0, 255]$

Extremely expensive even  
for low-resolution images

**Raster-order image generation**

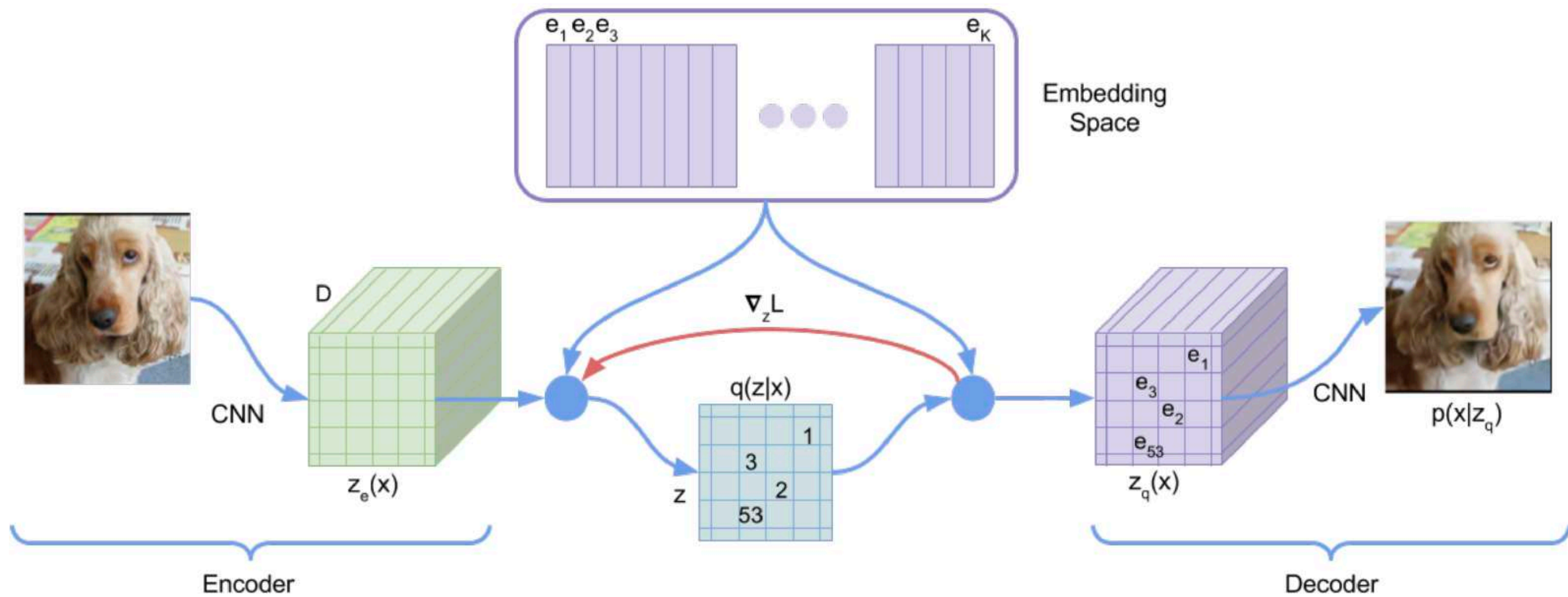


# Continuous Autoencoders

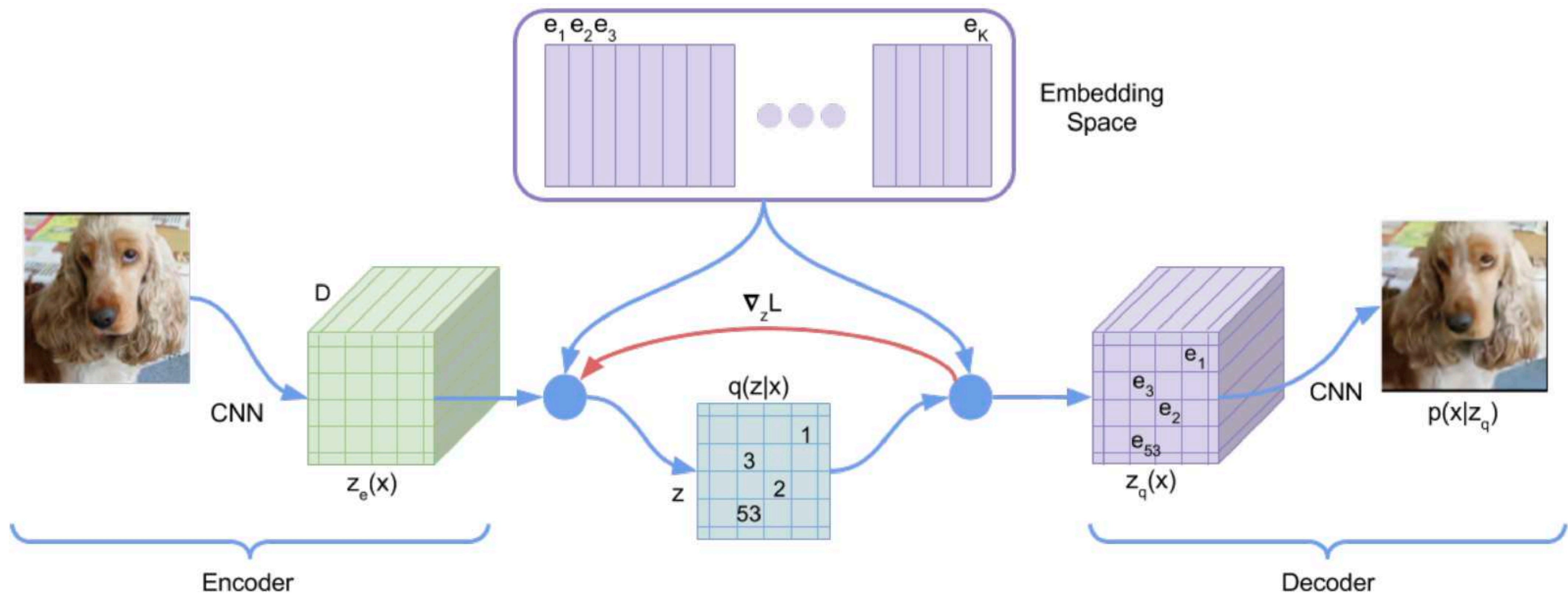


**Reconstruction loss:**  $L = \log p(\mathbf{x} \mid \mathbf{z}_e(x))$ , e.g., MSE.

# Vector Quantization Autoencoders



# Vector Quantization Autoencoders



**VQ-VAE loss:** 
$$L = \log p(\mathbf{x} \mid \mathbf{z}_q(x)) + \left\| \text{sg}[\mathbf{z}_e(x)] - e \right\|_2^2 + \beta \left\| \mathbf{z}_e(x) - \text{sg}[e] \right\|_2^2$$

# VQ-VAE follow-ups

**VQ-GAN** — VQ-VAE but add GAN and LPIPS losses for better reconstruction

## ViT-VQ-GAN

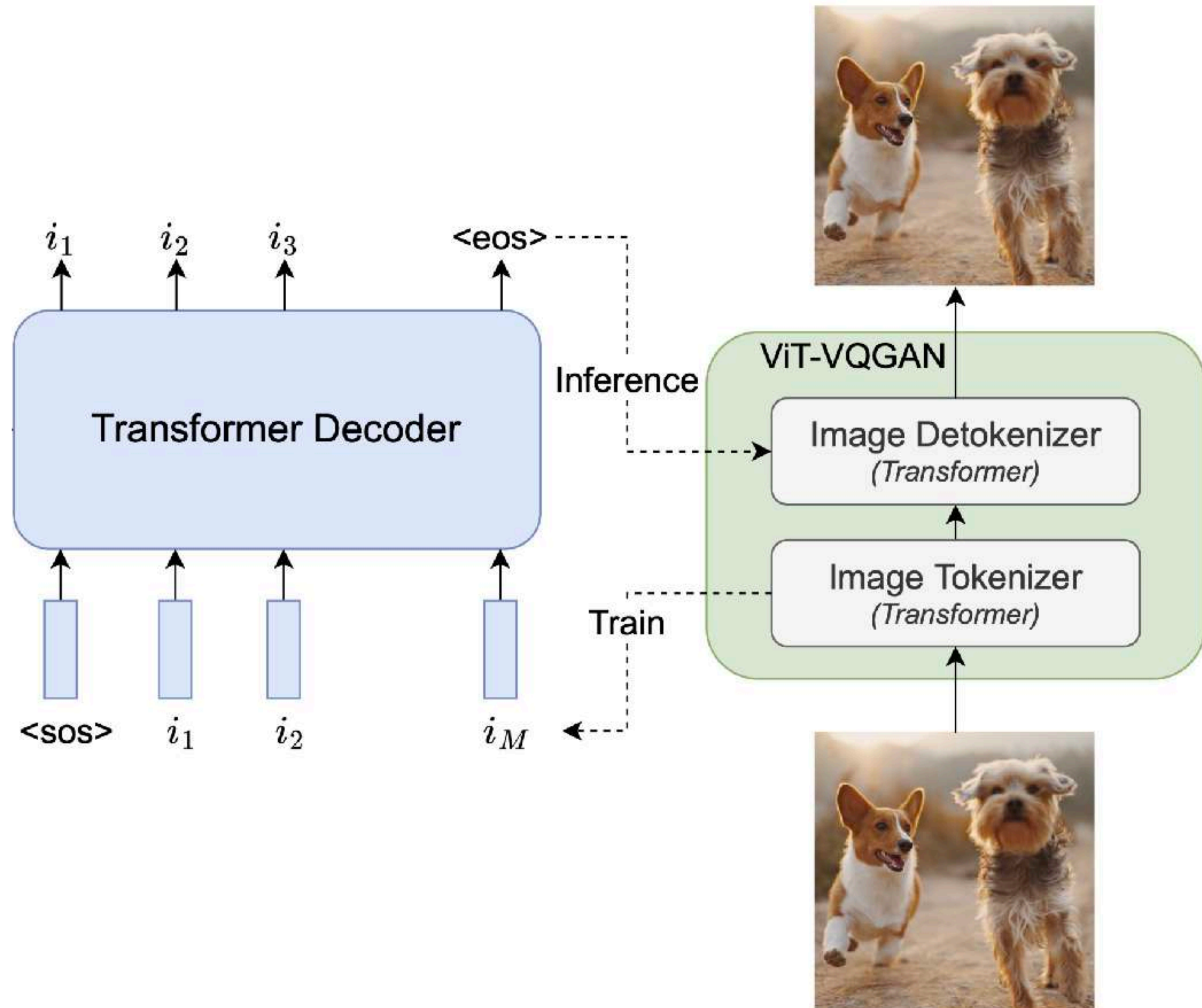
— Convolutional AE → ViT-AE (Not critical)

Address bad codebook utilization:

- Normalize latents and codebook vectors
- Reduce dimensionality of codebook vectors

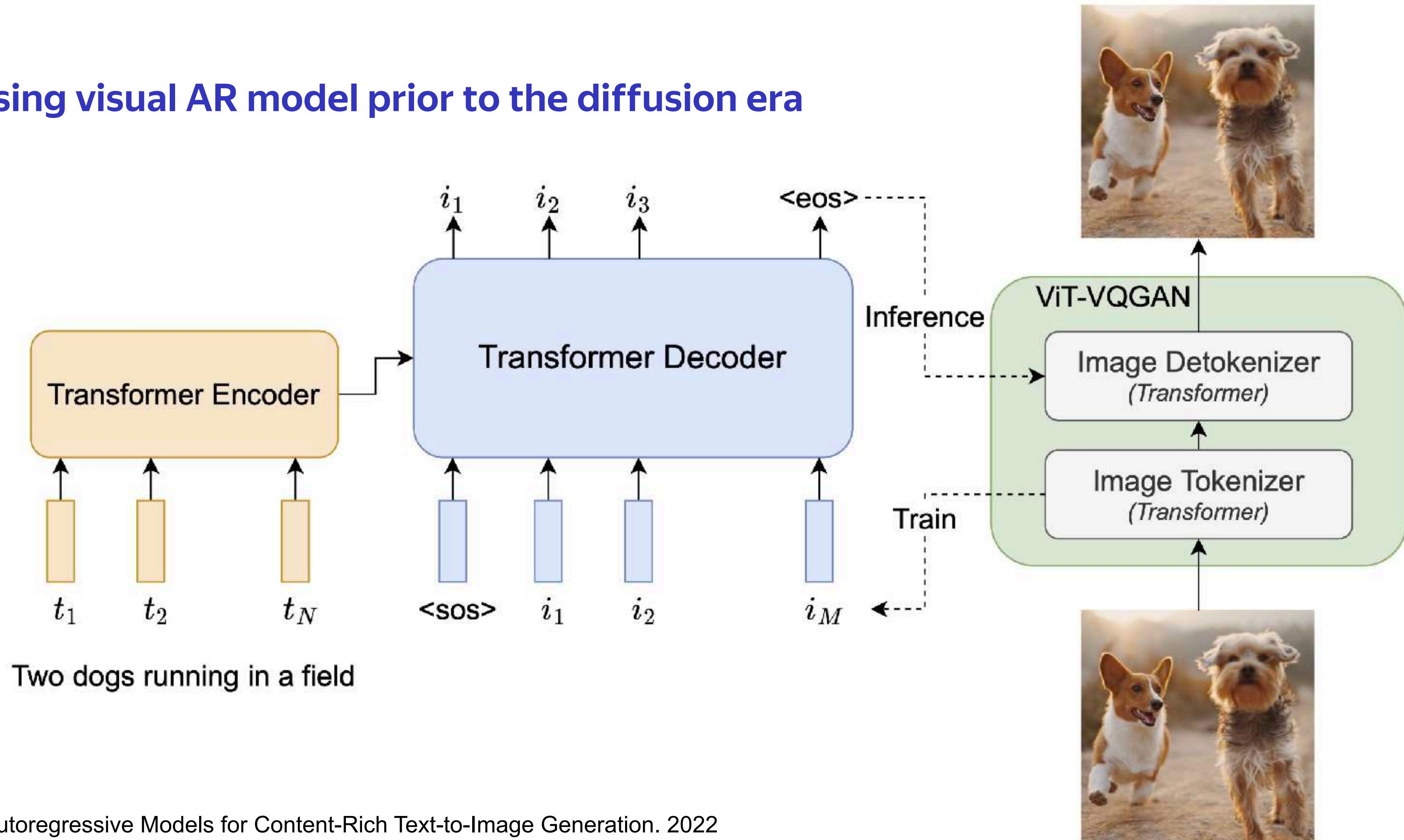
**In practice:** convolutional VQ-GAN + ViT-VQ-GAN tricks

# Image AR generation



# Text-to-image AR generation | Parti

Last promising visual AR model prior to the diffusion era



# Sampling with discrete models

All typical sampling techniques from LLMs are valid here

Top-k and Top-p (nucleus) sampling

Temperature in softmax

Beam search *but I didn't see it in practice*

There are more image specific techniques

**CFG:**  $\text{logits}_{cf\text{g}} = \text{logits}_{uncond} + \gamma \cdot (\text{logits}_{cond} - \text{logits}_{uncond})$

Gumbel-softmax averaging *quite specific thing, never mind*



A FEW

~~MOMENTS~~ LATER

YEARS

# LlamaGen

Using more recent LLM architectures

RMSNorm pre-normalization

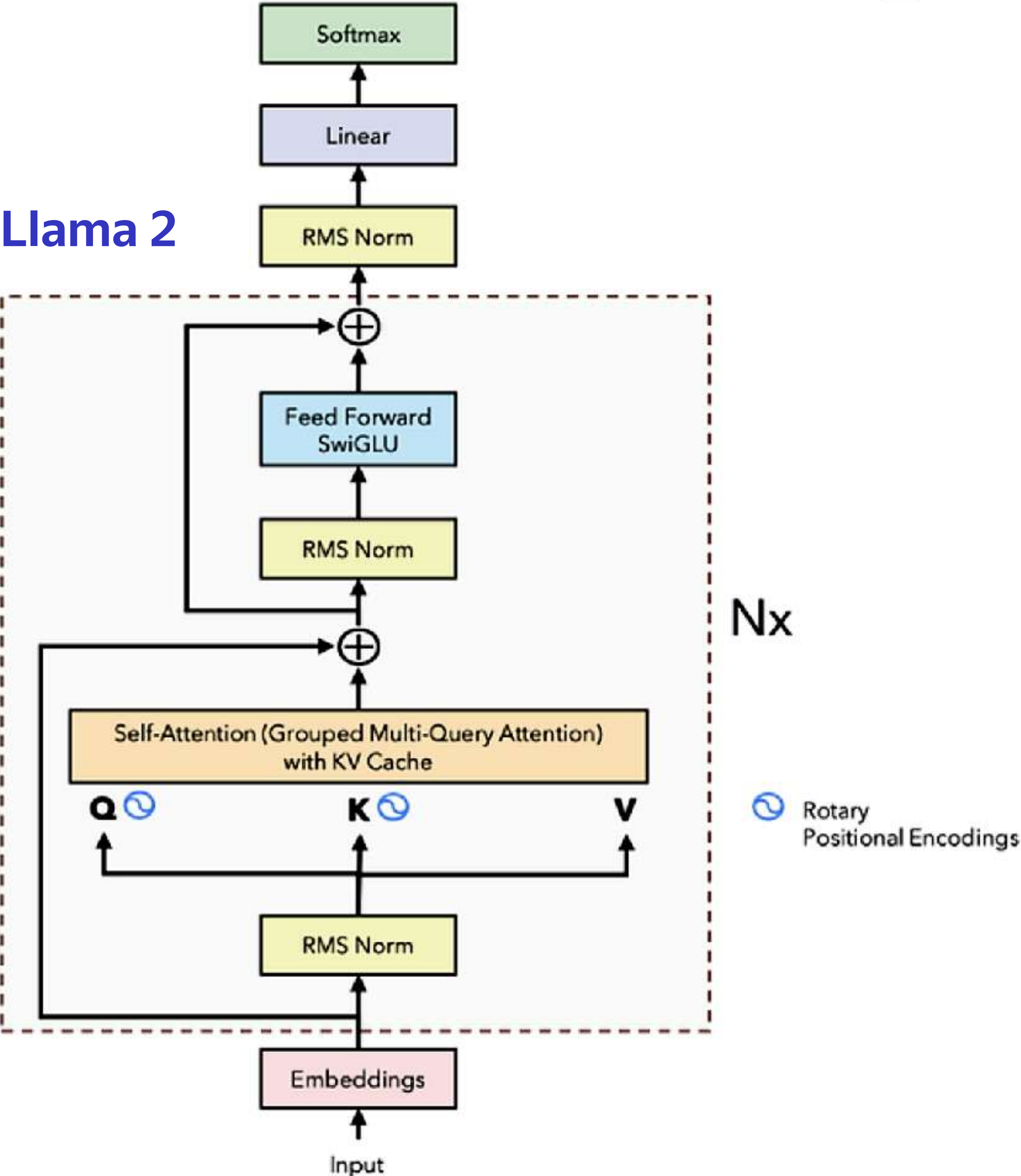
2D RoPE

In-context conditioning

SwiGLU

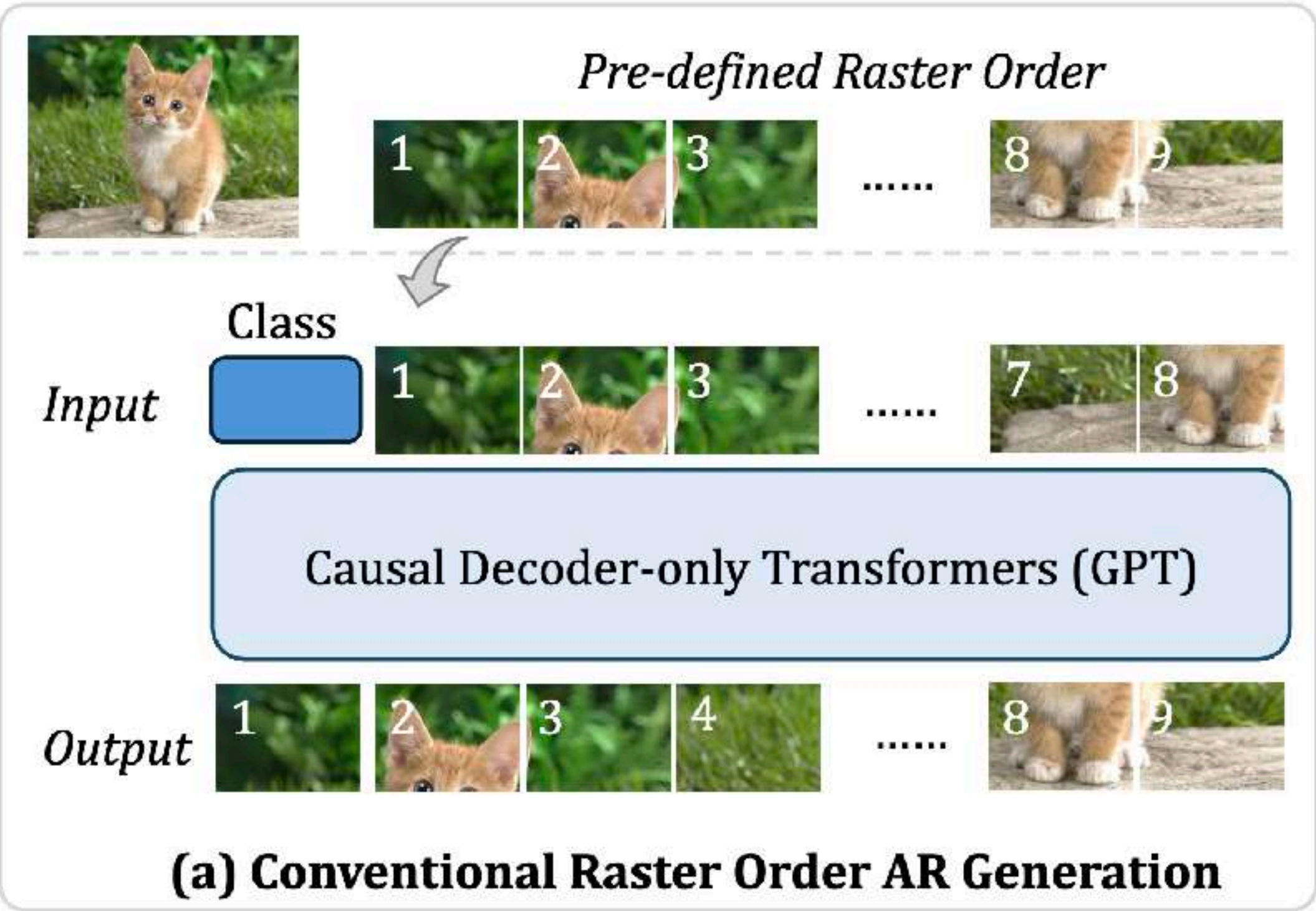
Moving closer to diffusion but no...

Llama 2



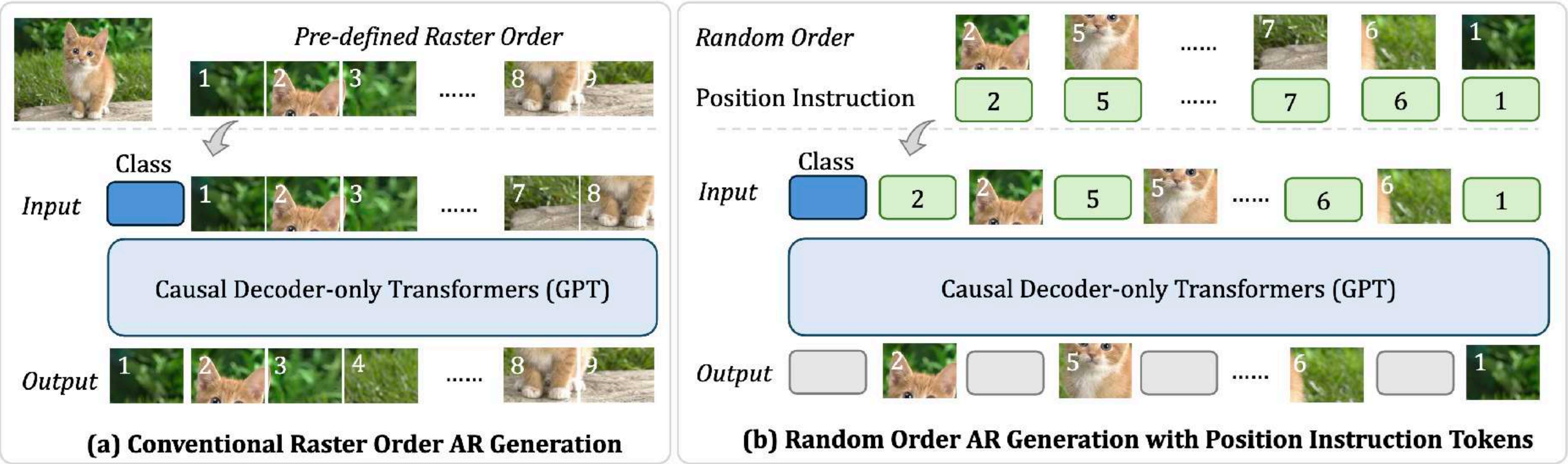
# Raster-order AR models

LlamaGen performs raster-order AR generation



# Any-order AR models

RandAR enables AR generation in any orders by introducing **Position Instruction Tokens**

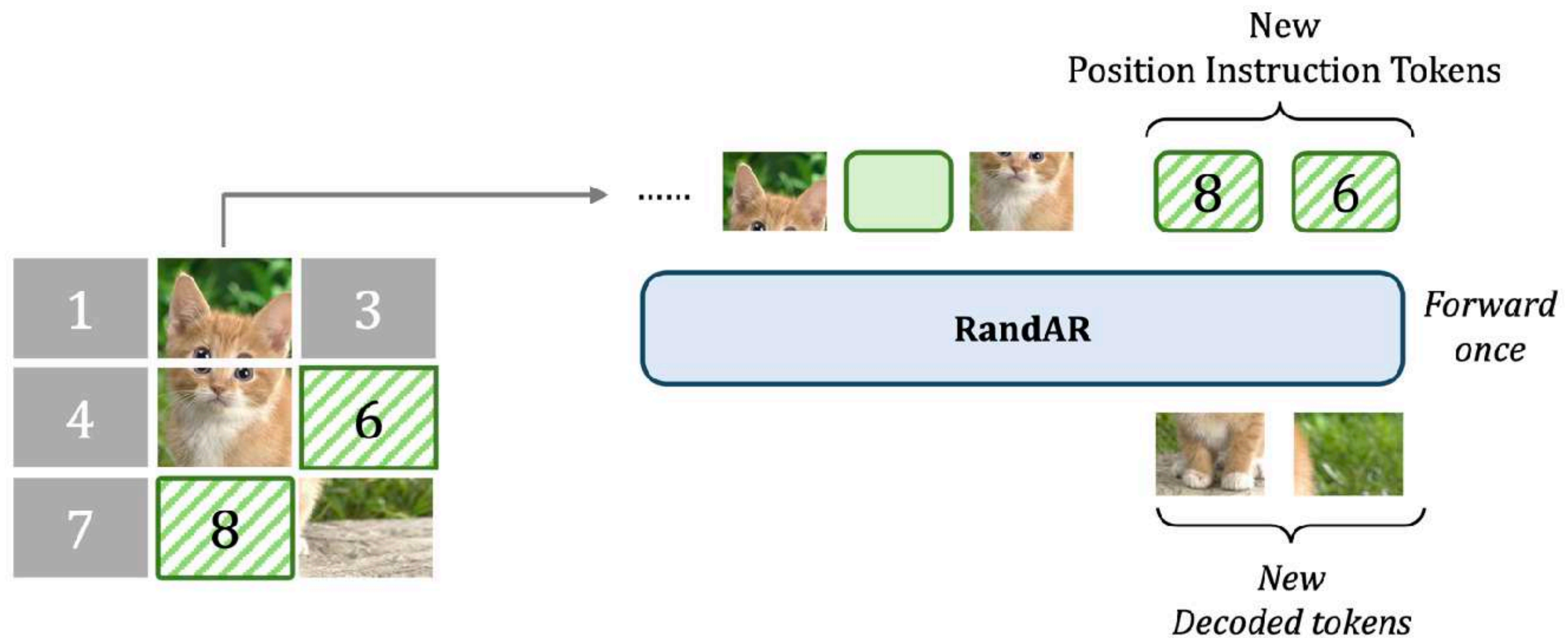


# Any-order AR models

**Does any order improve image quality? No, but...**

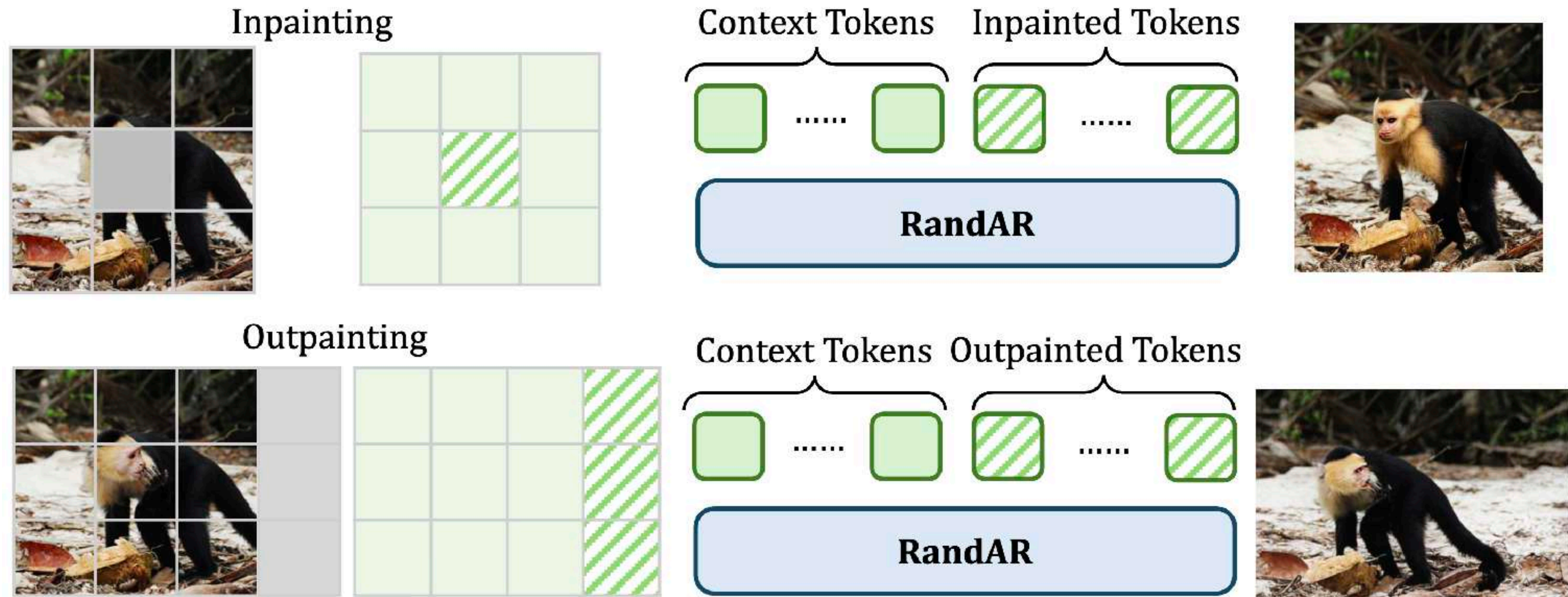
# Any-order AR models

Does any order improve image quality? No, but **parallel decoding**



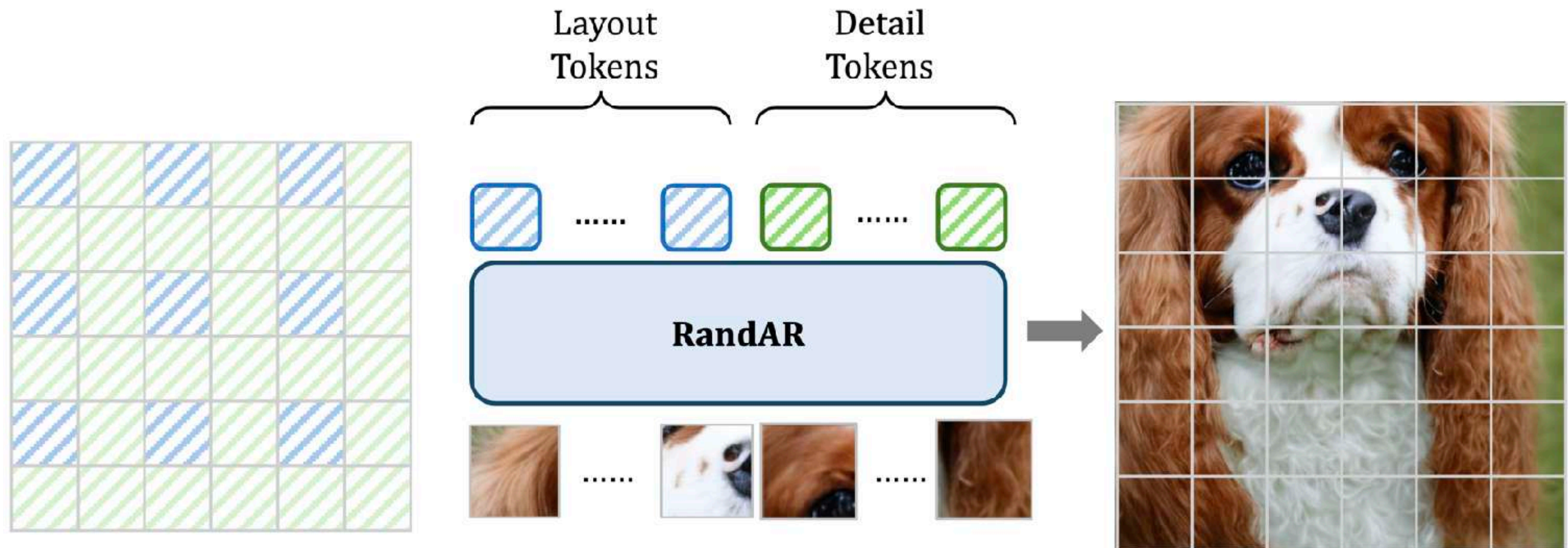
# Any-order AR models

Does any order improve image quality? No, but **inpainting & outpainting**



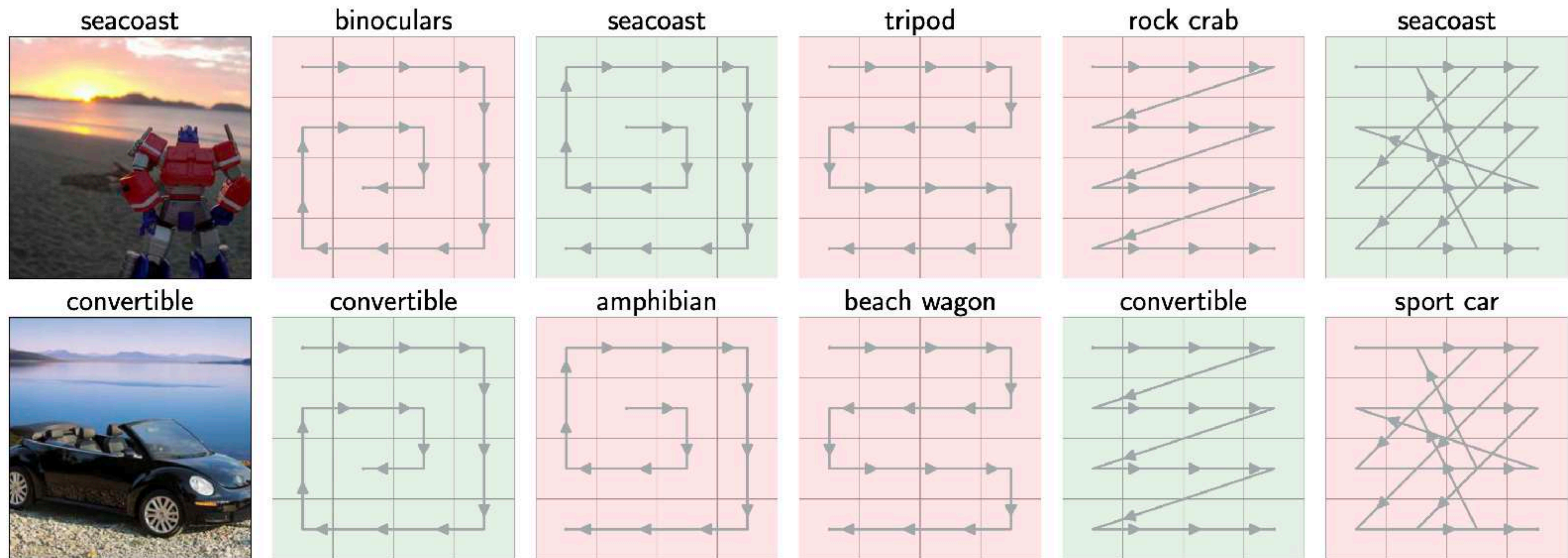
# Any-order AR models

Does any order improve image quality? No, but **zero-shot resolution extrapolation**



# Any-order AR models

Does any order improve image quality? No, but **better AR-based image classification**

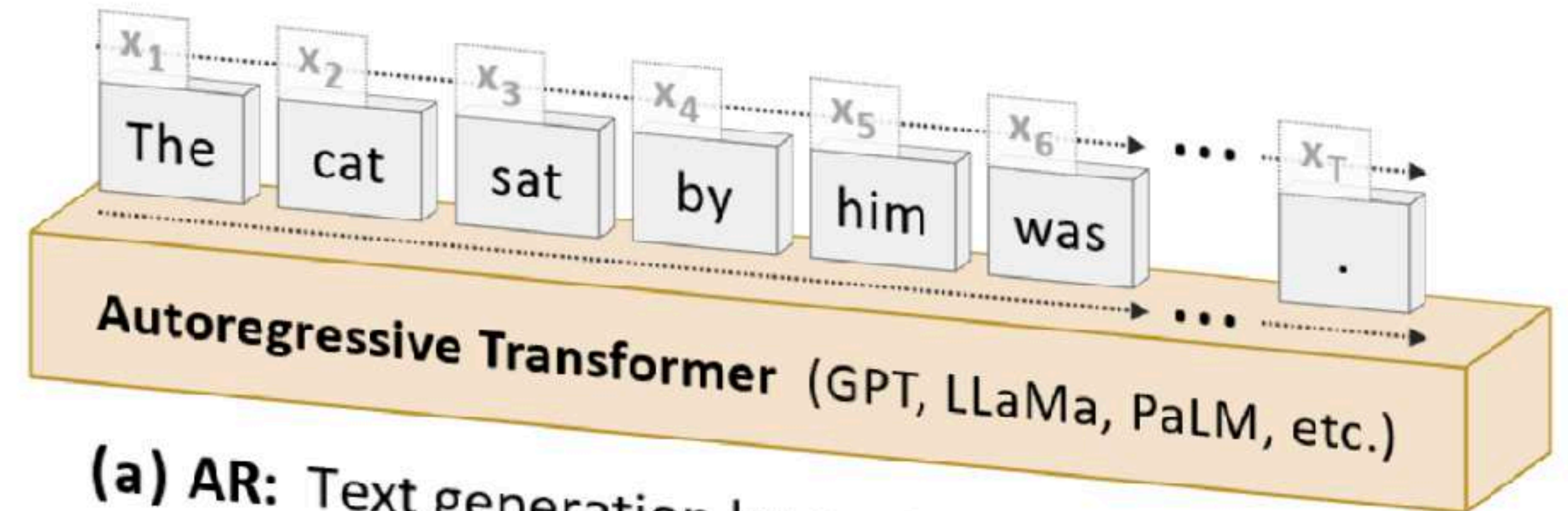


# Why GPT-like AR is bad for visual domains?

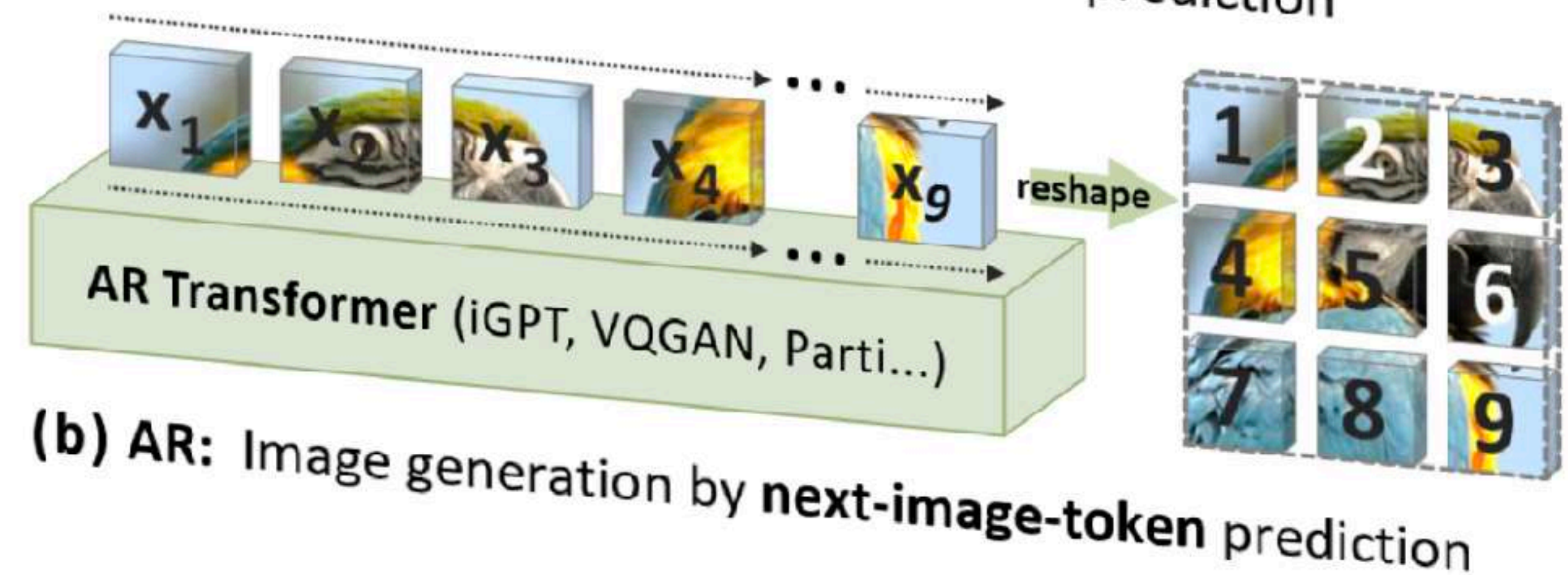
Next-token prediction is unnatural for visual data

Discrete representations are worse for both image reconstruction and generators

Low inference speed even in latent space



(a) AR: Text generation by **next-token** prediction

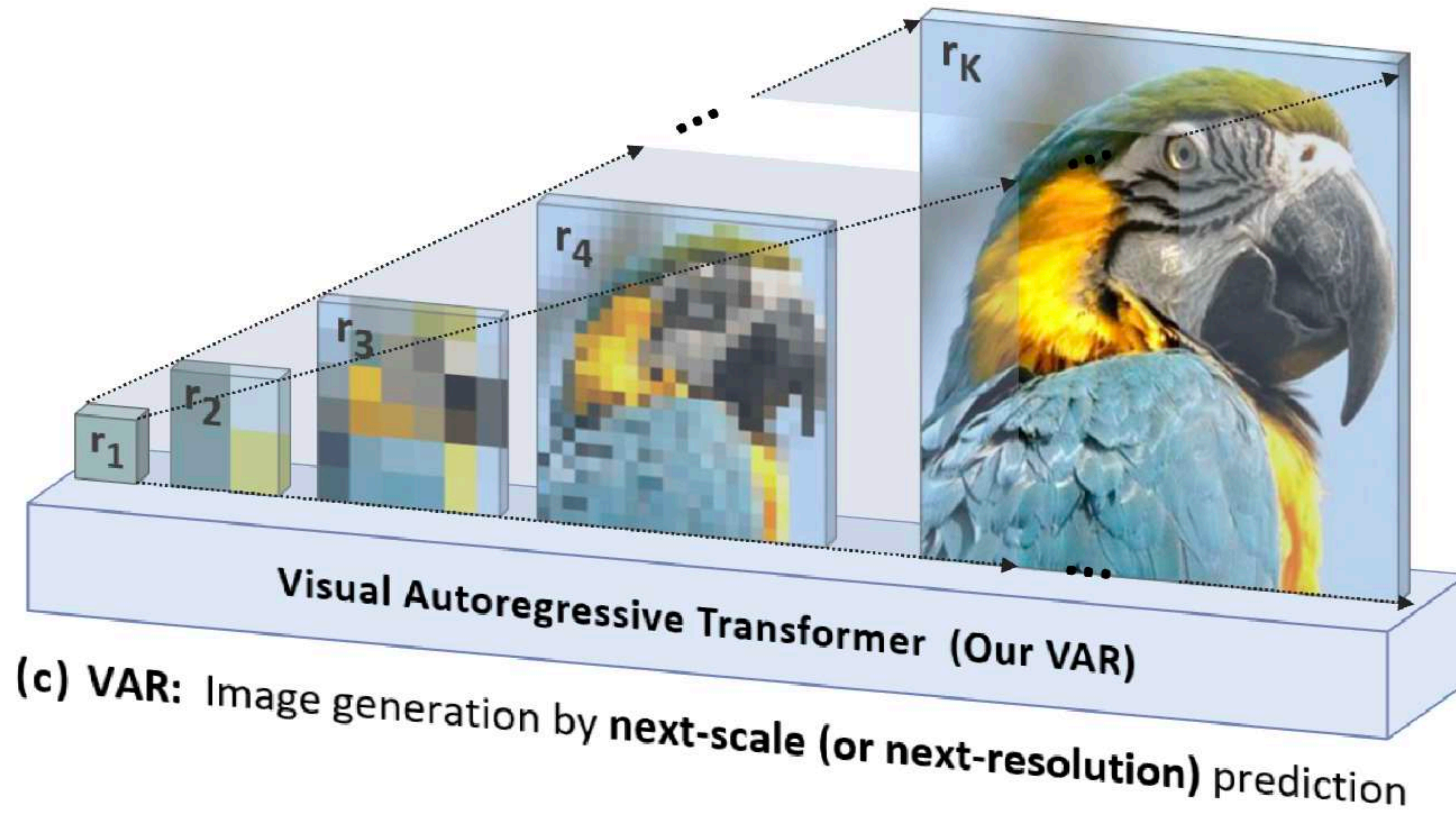


(b) AR: Image generation by **next-image-token** prediction

# Next-scale prediction AR models

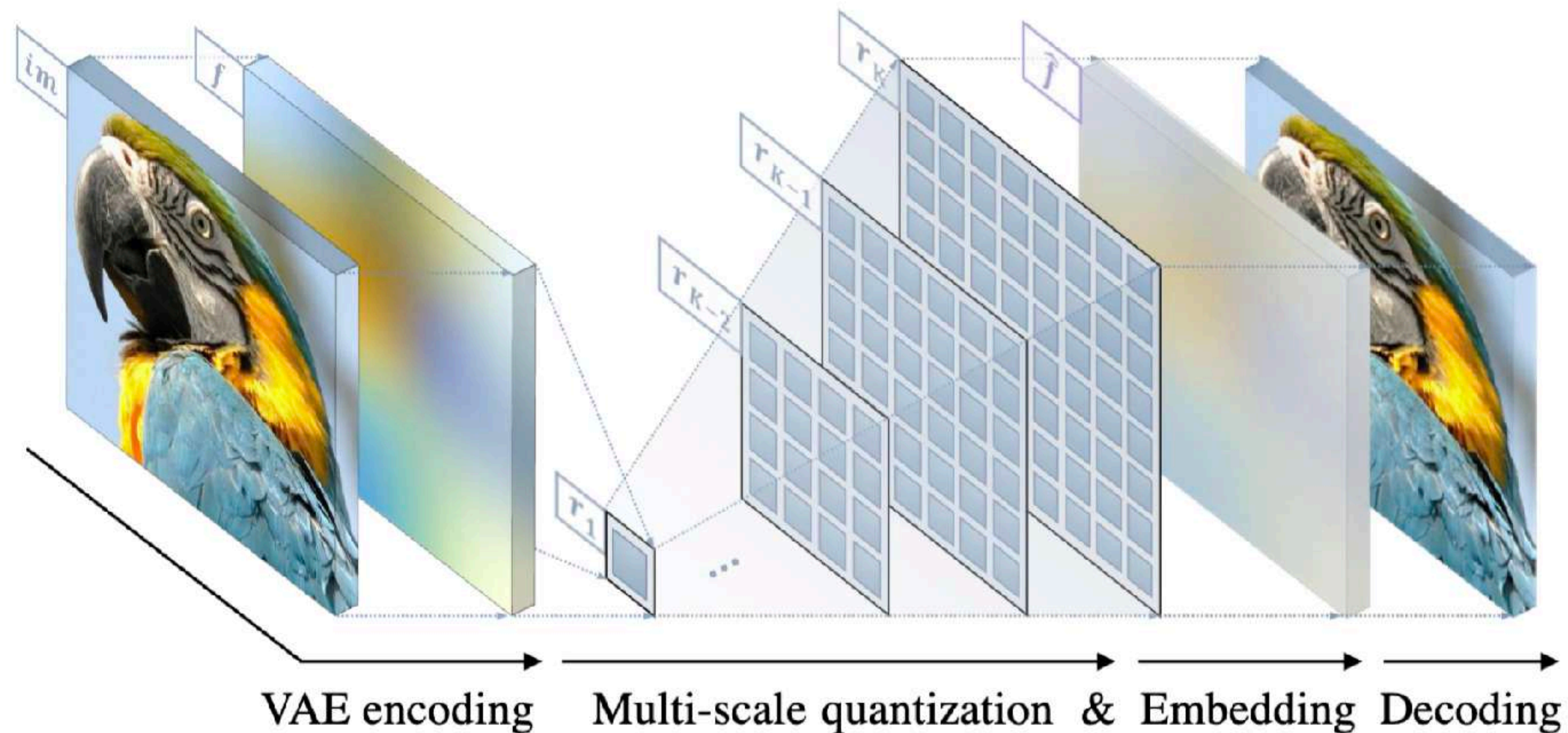


# Scale-wise Visual AR Generation

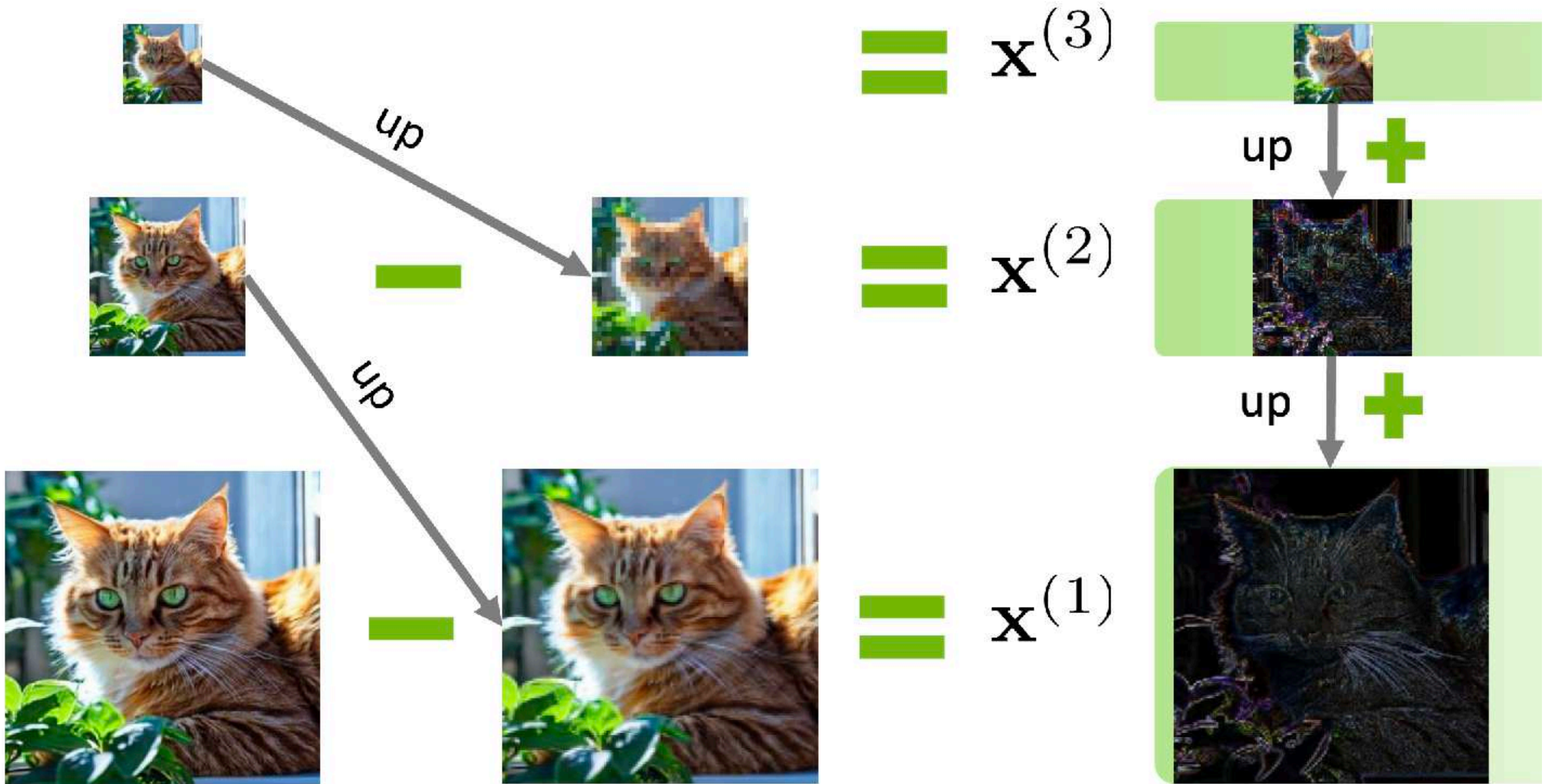


# Hierarchical multi-scale VQ-VAE

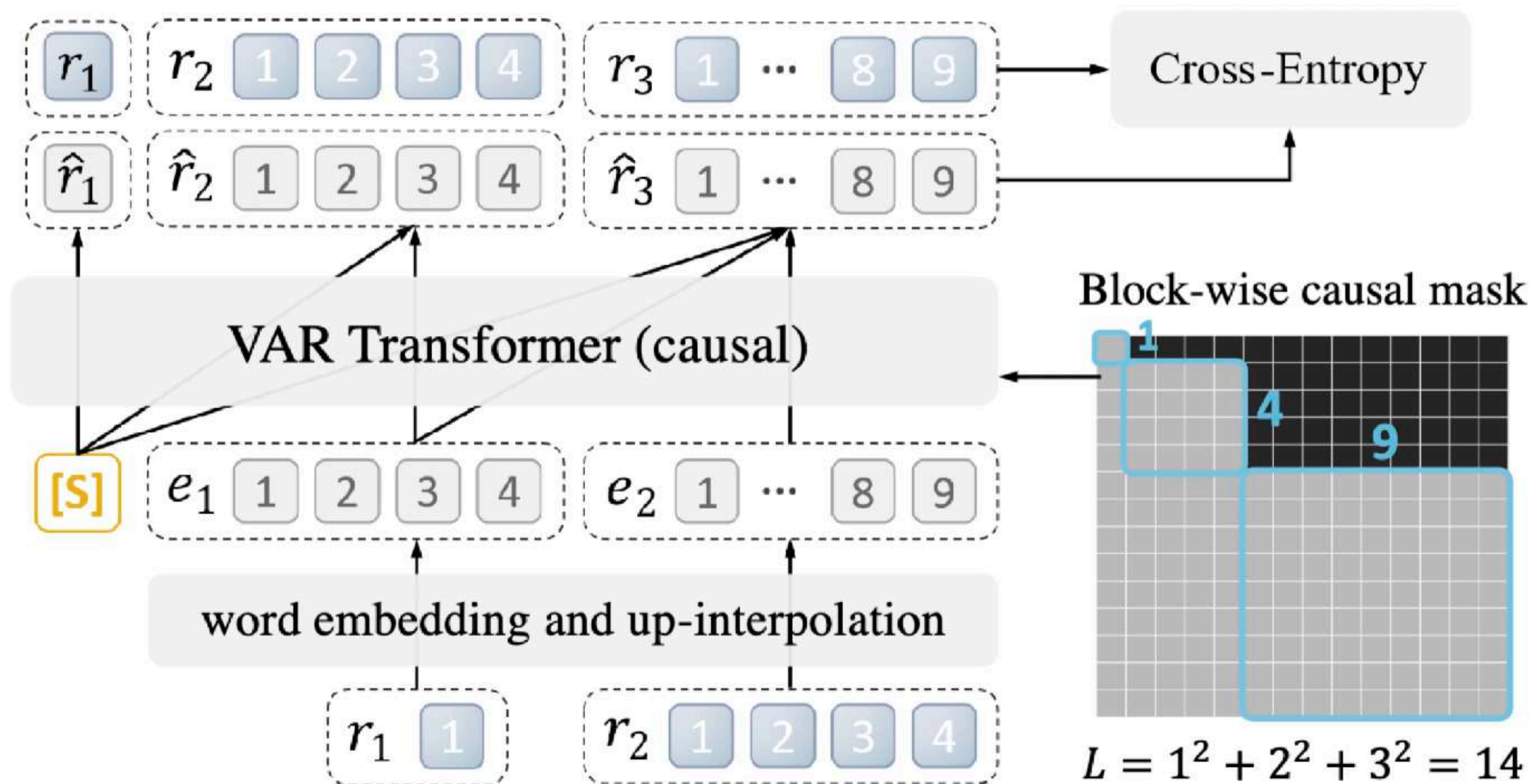
**Residual Quantization:**  $\hat{\mathbf{f}} = up(\mathbf{r}_1, 8\times) + up(\mathbf{r}_2, 4\times) + up(\mathbf{r}_3, 2\times) + \mathbf{r}_4$



# Laplacian decomposition

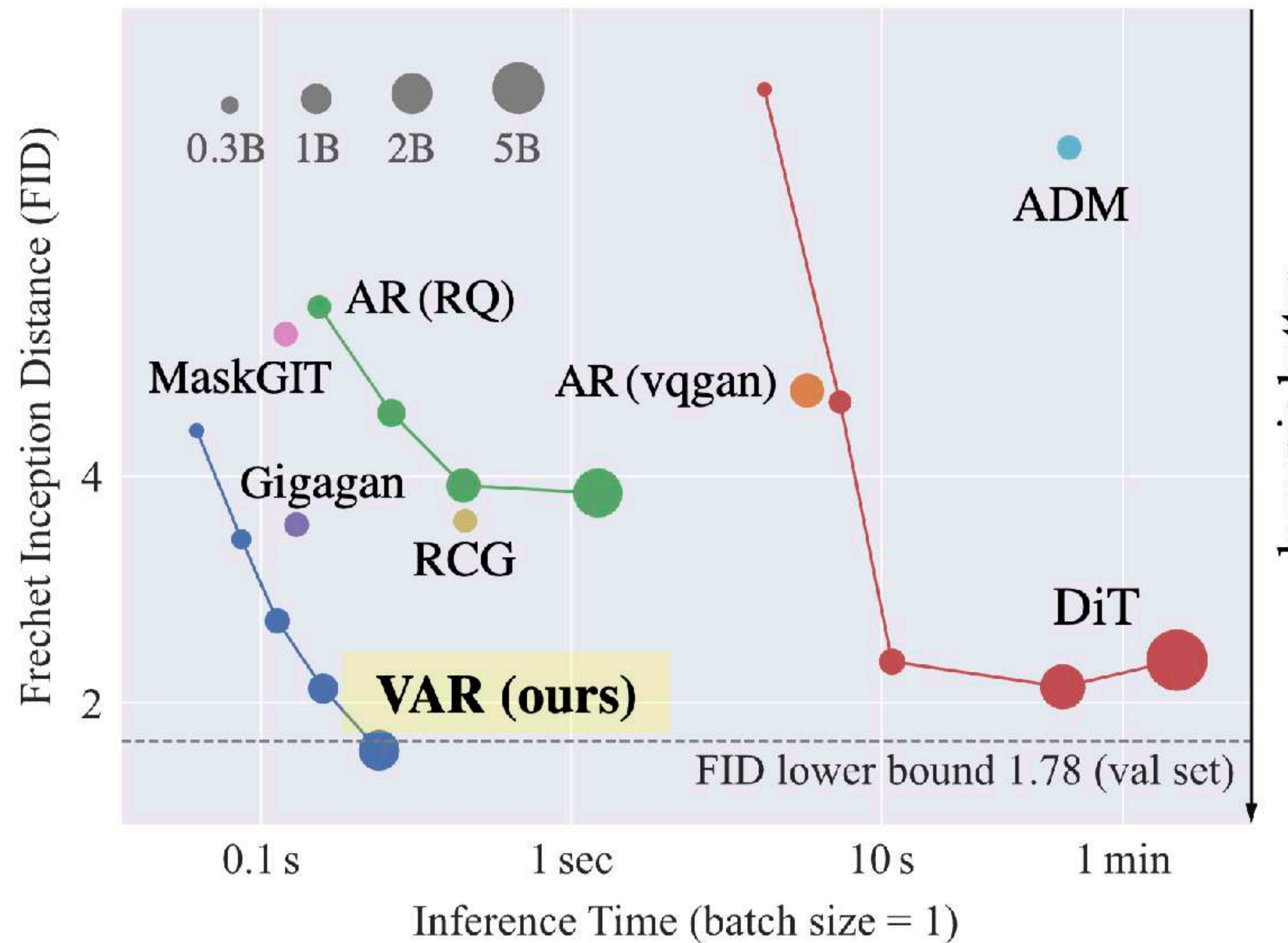


# VAR training



# VAR results

What about text-to-image generation?



# SWITTI: Designing Scale-Wise Transformers for Text-to-Image Synthesis

Anton Voronov<sup>1,2,3</sup>

Denis Kuznedelev<sup>1,4</sup>

Mikhail Khoroshikh<sup>6</sup>

Valentin Khrulkov<sup>1,5</sup>

Dmitry Baranchuk<sup>1</sup>

<sup>1</sup>Yandex Research

<sup>2</sup>HSE University

<sup>3</sup>MIPT

<sup>4</sup>Skoltech

<sup>5</sup>AIRI

<sup>6</sup>ITMO

<https://yandex-research.github.io/switti>



Figure 1. SWITTI produces high quality and aesthetic 1024×1024 image samples in around 0.5 seconds.

# Designing and Scaling T2I VAR

**Goal: design and train first T2I VAR**

## Project directions

### T2I VAR Architecture

*CLIP text conditioning  
via CA and AdaLN*

*Non-causal attention*

*“Sandwich” normalization  
for stability*

*Multi-scale RoPE*

### Large-scale training

*2.5B model*

*100M dataset*

*320 A100 GPUs*

*256 → 512 → 1024 stages*

*SFT post-training*

### Analysis

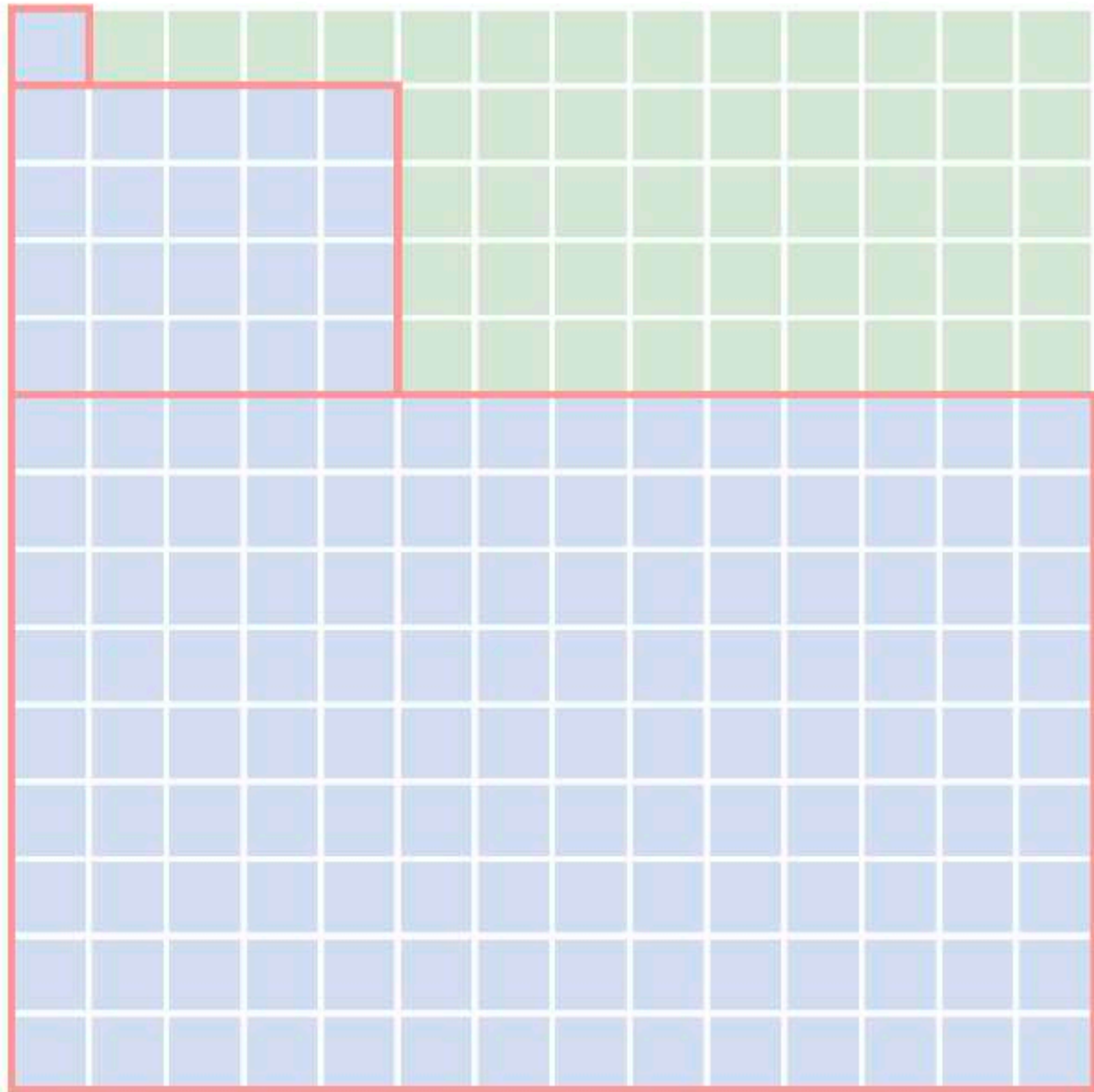
*Activation norms*

*Attention behavior*

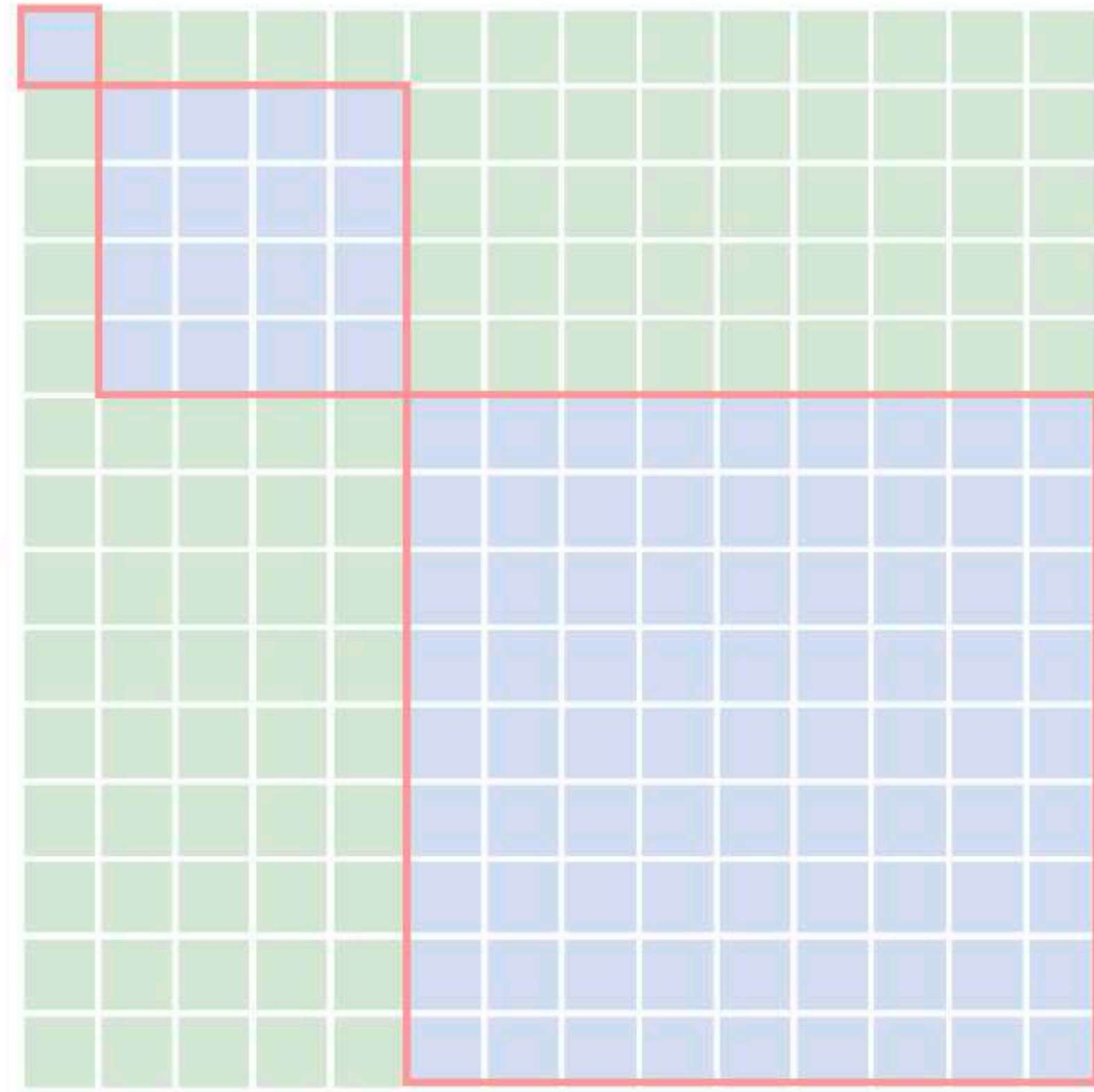
*Role of text-conditioning*

# Switti is no longer **explicitly** autoregressive

Block-wise causal mask (VAR)



Block-wise non-causal mask (Switti)



# Switti is **implicitly** autoregressive

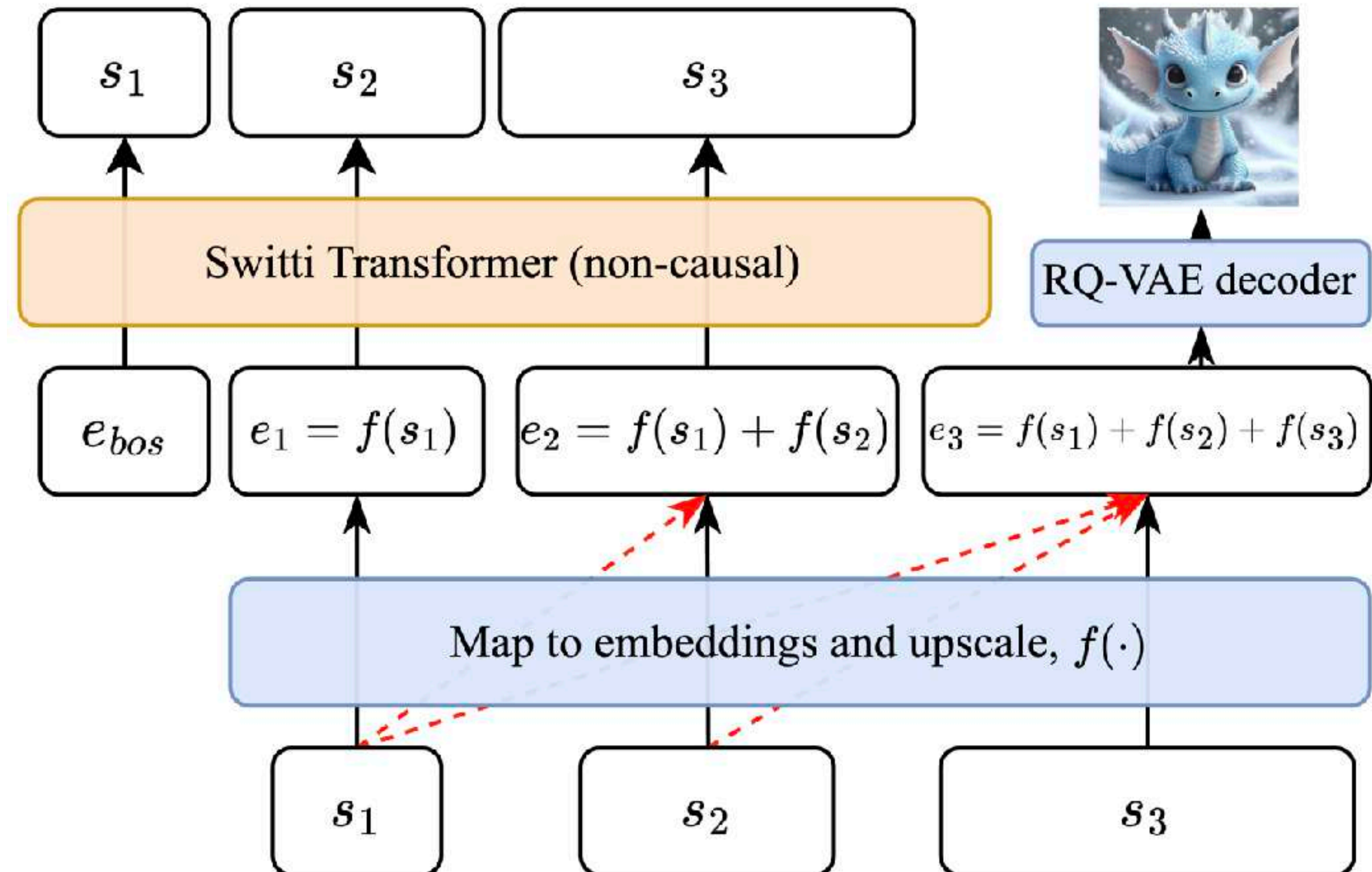
## Flexible training

- No need to process all scales at once

## More efficient inference

- No KV-cache (saves 2.4GB/image)
- Cheaper attention calculation (~21%)

## Slightly higher quality



# Effect of text conditioning

Weak text reliance  
at last scales



Turn off CFG



~32% speedup

Less defects



# VAR Summary

Highly natural visual inductive bias

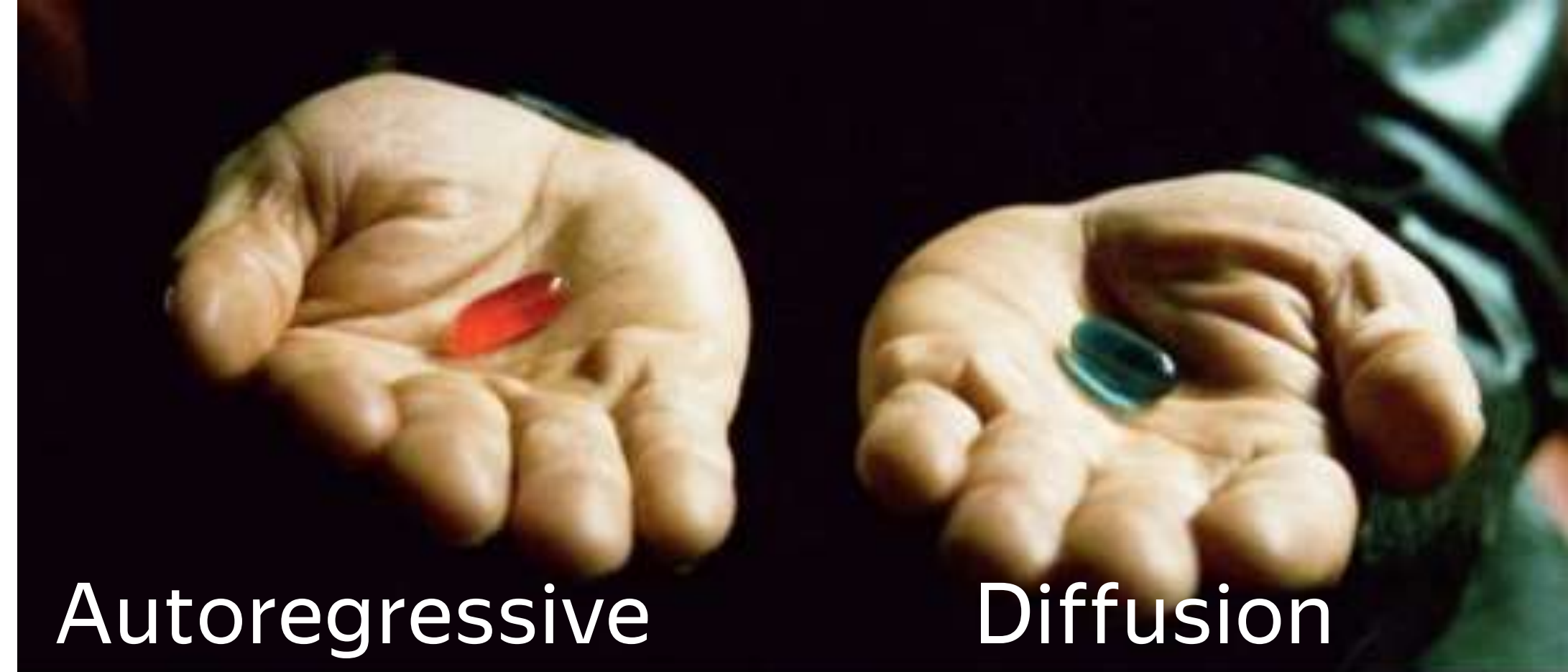
Much faster than DMs by design

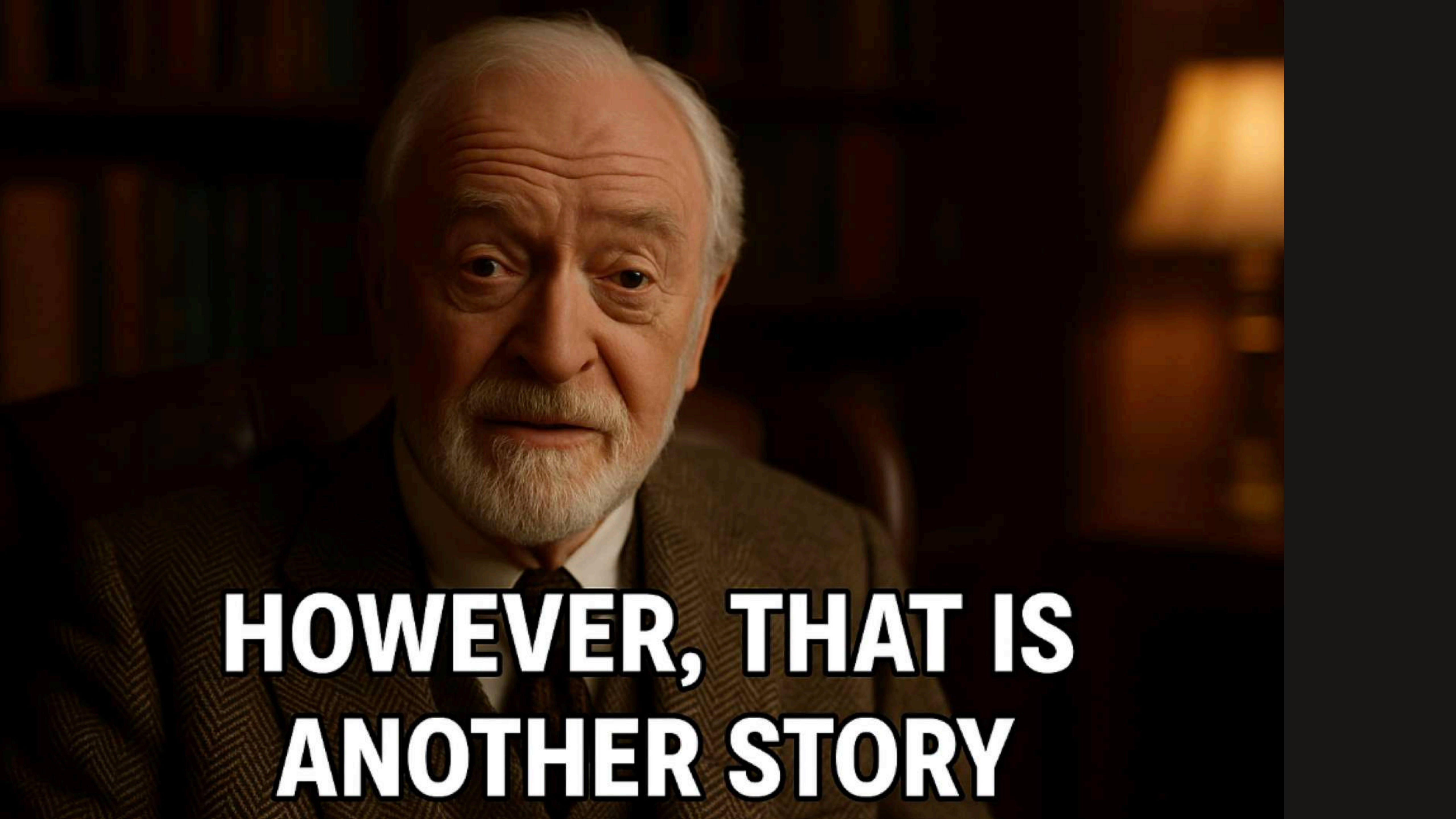
Still requires discrete tokenizers

Promising results but still needs much work to achieve top-tier DM performance



# Continuous AR models





**HOWEVER, THAT IS  
ANOTHER STORY**